SURYA HELIOPHYSICS FOUNDATION MODEL

Main Functions, Purposes, and Applications

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EXECUTIVE SUMMARY

Surya (Sanskrit for "Sun") is a 366-million parameter foundation model developed by IBM Research and NASA specifically for heliophysics and space weather prediction. It represents the first foundation model trained on full-resolution Solar Dynamics Observatory (SDO) data, enabling unprecedented capabilities in solar forecasting and analysis across multiple critical space weather tasks.

Key Innovation: Surya learns general-purpose solar representations through spatiotemporal transformers, allowing it to be fine-tuned for diverse downstream applications without task-specific model development.

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1. WHAT IS SURYA?

1.1 Technical Overview

Model Specifications:

- **Architecture:** Vision transformer with spectral gating and long-short range attention
- **Parameters:** 366 million trainable parameters
- **Training Data:** 9 years of full-resolution SDO observations (2011-2019)
- **Data Volume:** ~257 terabytes processed
- **Resolution:** Native 4096×4096 pixels, 12-minute cadence
- **Input Channels:** 13 channels from SDO's AIA (8 channels) and HMI (5 channels) instruments

Data Sources:

- **AIA (Atmospheric Imaging Assembly):** 8 extreme ultraviolet channels capturing solar atmosphere at different temperatures (94Å, 131Å, 171Å, 193Å, 211Å, 304Å, 335Å, 1600Å)
- **HMI (Helioseismic and Magnetic Imager):** 5 channels mapping magnetic field components and photosphere (line-of-sight magnetogram, vector magnetograms Bx/By/Bz, Doppler velocity)

1.2 Development Context

Developers: IBM Research in collaboration with NASA IMPACT and 8 research institutions:

- Development Seed
- Georgia State University
- Princeton University
- SETI Institute
- SouthWest Research Institute
- University of Alabama in Huntsville
- University of Colorado, Boulder
- Harvard-Smithsonian Center for Astrophysics

Release Date: August 20, 2024

Availability:

- Open-source on GitHub: https://github.com/NASA-IMPACT/Surya
- Model weights on HuggingFace: https://huggingface.co/nasa-ibm-ai4science
- Integration via IBM's TerraTorch library

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2. PRIMARY PURPOSES

2.1 Space Weather Forecasting

Problem Context:
Solar storms can:

- Endanger astronauts in space
- Knock satellites off orbit
- Disrupt GPS systems
- Cause power grid failures
- Interfere with long-distance communications

- Impact internet infrastructure

Current Limitation:
Solar flares travel at light speed—taking only 8 minutes to reach Earth. Traditional methods provide approximately 1-hour advance warning, insufficient for protective measures.

Surya's Solution:
Provides **2-hour advance warning** for solar flares (100% improvement over existing methods), giving critical additional time for:

- Satellite operators to place spacecraft in safe mode
- Power grid managers to stabilize distribution systems
- Aviation authorities to reroute polar flights
- Space agencies to protect astronauts

2.2 Scientific Discovery and Data Analysis

Challenge:
NASA's Solar Dynamics Observatory (SDO) has captured 15 years of continuous solar observations, generating petabytes of data. Manual analysis of this volume is impractical.

Surya's Capability:
Automates feature extraction and event detection from massive datasets, enabling researchers to:

- Identify patterns across years of observations
- Discover previously unnoticed solar phenomena
- Test hypotheses at unprecedented scale
- Accelerate scientific research cycles

Impact Statement from Dr. Kathy Reeves (Harvard-Smithsonian Center for Astrophysics):
"Pulling features and events out of petabytes of data is a laborious process and now we can automate it."

2.3 Foundation Model Paradigm for Space Physics

Innovation:
First demonstration that foundation model approaches (successful in NLP and computer vision) can be applied to space physics.

Advantage:
Single pre-trained model can be adapted for multiple tasks through fine-tuning, eliminating need to train specialized models from scratch for each application.

Broader Impact:
Establishes methodology for applying AI to other areas of space science and planetary observation.

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3. MAIN FUNCTIONS AND CAPABILITIES

3.1 Function 1: Solar Flare Forecasting

Purpose:
Predict M-class and X-class solar flares up to 24 hours in advance.

Technical Approach:

- Binary classification (flare/no-flare)
- Analysis of active region evolution
- Pattern recognition in magnetic field topology
- Visual and magnetic signatures integration

Performance Metrics:

- **True Skill Statistic (TSS):** 0.436
- **Improvement over baseline:** 16% better than previous state-of-the-art (AlexNet: TSS = 0.358)
- **Lead time:** 2 hours (vs 1 hour for traditional methods)

Real-World Impact:

- Early warning for satellite operators
- Protection protocols for astronauts
- Power grid preparedness
- Aviation route planning (polar flights)

Scientific Significance:
Provides longest lead time currently available for solar flare prediction, critical for space weather mitigation strategies.

3.2 Function 2: Active Region Segmentation

Purpose:
Automatically identify and segment solar active regions (ARs) and polarity inversion lines from magnetogram data.

Importance:
Active regions are the source of all space weather phenomena:

- Solar flares originate in active regions
- Coronal mass ejections (CMEs) emerge from AR magnetic structures
- Solar wind acceleration begins near active regions

Technical Approach:

- Pixel-wise semantic segmentation
- Boundary detection and localization
- Magnetic polarity identification
- Multi-scale feature extraction

Performance Metrics:

- **Intersection over Union (IoU):** 0.768
- **Improvement over baseline:** 12% better than UNet (IoU = 0.688)

Applications:

1. **Automated monitoring:** Real-time AR tracking across solar rotation
2. **Flare prediction input:** AR characteristics feed into forecasting models
3. **Scientific research:** Statistical studies of AR evolution and properties
4. **Mission planning:** Identifying high-activity regions for targeted observations

Quote from Dr. Spiros Kasapis (Princeton University):
"These [active regions] are the regions that produce all the magnetic particles we're interested in. This is essentially the very beginning of our relationship with the Sun."

3.3 Function 3: Extreme Ultraviolet (EUV) Spectra Prediction

Purpose:
Predict 1343-dimensional EUV spectral irradiance across 5-35 nm wavelength range.

Scientific Context:
EUV radiation causes Earth's upper atmosphere to:

- Heat and expand during solar storms
- Force satellites to fall from orbit (atmospheric drag increases)
- Disrupt radio communications
- Impact satellite lifetime and orbital decay rates

Technical Approach:

- High-dimensional regression (1343 output dimensions)
- Spectral pattern learning
- Multi-wavelength correlation modeling
- Temporal evolution prediction

Performance Metrics:

- **Mean Absolute Percentage Error (MAPE):** 1.48%
- Captures spectral shape with high fidelity
- Predicts irradiance variations accurately

Applications:

1. **Satellite drag modeling:** Predict atmospheric density for orbital calculations
2. **Communication forecasting:** Anticipate ionospheric disturbances
3. **Solar cycle studies:** Track long-term EUV variability
4. **Climate modeling:** EUV influence on upper atmosphere chemistry

Operational Importance:
Space agencies use EUV predictions for:

- Satellite collision avoidance maneuvers
- Re-entry calculations for deorbiting spacecraft
- International Space Station altitude maintenance planning

3.4 Function 4: Solar Wind Speed Forecasting

Purpose:
Predict solar wind speed at Earth's L1 point (1.5 million km from Earth) with 4-day lead time.

Critical Need:
Solar wind variations drive geomagnetic storms that can:

- Damage electrical transformers
- Induce currents in pipelines
- Disrupt radio communications
- Create hazardous radiation environments for astronauts
- Affect GPS accuracy

Technical Approach:

- Time-series regression
- Corona-to-heliosphere modeling
- Multi-day prediction horizon
- Speed range: 300-800 km/s

Performance Metrics:

- **Root Mean Square Error (RMSE):** 75.92 km/s
- 4-day advance prediction
- Captures slow and fast wind streams
- Identifies high-speed stream arrivals

Applications:

1. **Geomagnetic storm forecasting:** Predict Kp index and storm severity
2. **Power grid protection:** Pre-position resources, adjust load distribution
3. **Satellite operations:** Plan orbit-raising maneuvers during quiet periods
4. **Auroral predictions:** Tourist and scientific observation planning

Scientific Value:
Understanding solar wind acceleration and propagation remains a fundamental question in heliophysics. Surya's predictions provide insights into:

- Corona-heliosphere connection
- Fast/slow wind sources
- Stream interaction regions
- Magnetic structure evolution

Quote from Dr. Andrés Muñoz-Jaramillo (SouthWest Research Institute):
"Knowing what the solar wind will be when it rams our magnetic shield, or interacts with it, brings us one step closer to understanding its effects on our power distribution systems."

4. KEY TECHNICAL INNOVATIONS

4.1 Architecture Design

Long-Short Attention Mechanism:

- **Local attention:** Captures fine-scale spatial dependencies within image patches
- **Global attention:** Models long-range correlations across entire solar disk
- **Benefit:** Handles 4096×4096 resolution (10× higher than typical vision models)

Spectral Gating:

- Frequency-domain filtering with learnable complex weights
- Adaptive re-weighting of spectral components
- **Advantage:** 5% reduction in memory costs, improved noise filtering

Why This Matters:
Most vision transformers struggle with high-resolution scientific imagery. Surya's architecture enables processing of full-resolution SDO data without downsampling, preserving critical small-scale features.

4.2 Training Strategy

Two-Phase Approach:

Phase 1: One-step ahead forecasting

- Duration: 160,000 training steps
- Hardware: 128 GPUs
- Objective: Learn to predict solar state 1 hour in the future
- Purpose: Model learns fundamental solar physics and evolution

Phase 2: Autoregressive rollout tuning

- Extends prediction horizons to 2-5 hours
- Refines multi-step temporal consistency
- Optimizes for downstream task performance

Self-Supervised Learning: Model learns without manual labels by predicting future solar states, discovering relevant patterns automatically.

Quote from Johannes Schmude (IBM AI Researcher):
"It's remarkable that the model performed better when we let it figure out the rotation on its own" [referring to the Sun's differential rotation]

4.3 Data Processing Pipeline

Challenges Addressed:

1. **Temporal alignment:** Different instruments have different cadences
2. **Spatial registration:** Ensuring all channels aligned to same solar coordinates
3. **Calibration:** Correcting for instrument degradation over 9 years
4. **Quality control:** Filtering corrupted or missing data

Pipeline Output:

- Uniform 12-minute cadence across all 13 channels
- Co-registered 0.6 arcsec/pixel spatial resolution
- Solar north aligned coordinate system
- Consistent radiometric calibration

Impact:
Creates the first ML-ready, physics-consistent, multi-instrument solar dataset at full resolution.

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5. REAL-WORLD APPLICATIONS

5.1 Space Mission Protection

Use Case: Protecting astronauts and spacecraft from radiation

Implementation:

1. Surya predicts solar flare 2 hours in advance
2. Mission control receives automated alert
3. **Astronaut response:**
    - Cancel spacewalks
    - Move to shielded areas of spacecraft
    - Monitor radiation dosimeters
4. **Satellite response:**
    - Enter safe mode (turn off sensitive electronics)
    - Reorient to minimize exposed surface area
    - Postpone thruster burns or antenna deployments

Example Scenario:
International Space Station crew receives 2-hour warning before X-class flare, allowing them to:

- Complete critical repairs early
- Shelter during peak radiation
- Resume operations after all-clear

Quantifiable Benefit:
Reducing astronaut radiation exposure by even 10% could prevent significant long-term health risks on Mars missions.

5.2 Power Grid Stability

Use Case: Protecting electrical infrastructure from geomagnetic storms

Implementation:

1. Surya predicts solar wind speed increase (4-day lead time)
2. Forecast indicates high-speed stream arrival
3. **Power grid operators:**
    - Pre-position repair crews in high-risk areas
    - Adjust transformer loading (reduce capacity)
    - Stage backup equipment
    - Coordinate with adjacent grid regions
4. Storm arrives as predicted
5. Grid remains stable with minimal disruption

Historical Context:

- **1989 Quebec blackout:** Geomagnetic storm caused 9-hour power outage affecting 6 million people
- **Cost:** Estimated $2 billion in damages
- **Prevention potential:** 4-day warning would have enabled protective measures

Economic Impact:
U.S. Department of Homeland Security estimates a major space weather event could cause $1-2 trillion in damages. Early warning systems like Surya could reduce losses by 60-80%.

5.3 Satellite Operations and Orbital Predictions

Use Case: Managing satellite constellations during solar activity

Problem:
EUV radiation heats upper atmosphere, increasing drag on low-Earth orbit satellites.

Surya's Role:

1. Predicts EUV irradiance variations
2. Forecasts atmospheric density changes
3. Enables accurate orbital decay calculations

Operational Applications:

- **Starlink (SpaceX):** Optimize satellite positioning, avoid collisions
- **International Space Station:** Plan altitude-raising maneuvers
- **Space debris tracking:** Update trajectory predictions

Case Study - February 2022:
SpaceX lost 40 Starlink satellites ($100M) due to unexpected atmospheric expansion from geomagnetic storm. Surya-type predictions could have enabled:

- Delayed launch by 48 hours
- Different deployment altitude
- Saved entire satellite batch

5.4 Aviation Safety

Use Case: Polar flight route management

Problem:
High-latitude flights experience:

- Increased radiation exposure during solar storms
- Radio communication blackouts
- GPS navigation disruption

Implementation:

1. Surya predicts solar flare 2 hours in advance

2. Airlines receive automated alerts
3. **Actions taken:**
    - Reroute polar flights to lower latitudes
    - Delay departures
    - Increase crew rotation (radiation limits)
    - Use backup navigation systems

Airlines Affected:

- United Airlines: ~20 polar routes
- Air Canada: Multiple transpolar services
- Lufthansa: Frankfurt-Los Angeles polar route

Regulatory Context:
FAA and EASA require radiation monitoring for aircrew. Surya predictions help airlines comply while maintaining schedules.

5.5 GPS and Navigation Systems

Use Case: Maintaining positioning accuracy during solar storms

Impact of Solar Activity on GPS:

- Ionospheric scintillation (signal fluctuation)
- Timing errors from atmospheric delays
- Loss of signal lock
- Reduced accuracy from meters to tens of meters

Surya's Contribution:

1. Predicts EUV irradiance changes
2. Forecasts ionospheric disturbances
3. Enables GPS operators to:
    - Issue accuracy warnings to users
    - Adjust satellite orbits
    - Implement error correction algorithms
    - Switch to backup positioning systems

Critical Users:

- Military operations (precision weapons)

- Autonomous vehicles
- Precision agriculture
- Emergency services
- Financial systems (GPS time-stamping for transactions)

5.6 Scientific Research Applications

Application 1: Solar Cycle Studies

Surya enables analysis of:

- Active region emergence patterns across solar cycles
- Magnetic field evolution over years
- Flare frequency variations with solar activity
- EUV irradiance trends

Application 2: Coronal Heating Problem

One of heliophysics' greatest mysteries: Why is the Sun's corona millions of degrees hotter than its surface?

Surya contributions:

- Statistical analysis of thousands of active regions
- Correlation between magnetic structures and heating
- Identification of energy dissipation patterns

Application 3: Space Weather Model Validation

Researchers use Surya predictions to:

- Test physics-based models
- Identify model deficiencies
- Improve parametrizations
- Benchmark new forecasting approaches

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6. COMPARISON WITH EXISTING APPROACHES

6.1 vs. Traditional Physics-Based Models

Aspect	Traditional Models	Surya
Development time	Years of expert coding	Weeks of training
Computational cost	Hours per forecast	Minutes per forecast
Adaptability	Requires code rewrite	Fine-tune existing model
Data requirements	Specific model inputs	General SDO observations
Performance	Good for known physics	Better for complex patterns
Interpretability	High (physics equations)	Lower (learned features)

Best Use Case for Each:

- **Physics models:** When physical mechanisms are well-understood
- **Surya:** When patterns are complex or physics is uncertain

Complementary Approach:
Many researchers advocate for hybrid models combining physics-based constraints with Surya's learned representations.

6.2 vs. Task-Specific Machine Learning Models

Traditional ML Approach:

1. Train separate model for each task
2. Requires task-specific data collection
3. Cannot transfer knowledge between tasks
4. Limited by training data quality and quantity

Surya's Foundation Model Approach:

1. Single pre-trained model for all tasks
2. Fine-tune on small task-specific datasets
3. Transfer learning enables better performance

4. Leverages full SDO dataset for general understanding

Efficiency Comparison:

Metric	Traditional ML	Surya
Training data needed	10,000+ samples per task	1,000+ samples (fine-tuning)
Training time	Days-weeks per task	Hours per task (fine-tuning)
Model size	10-100MB per task	1.4GB shared
Performance	Task-dependent	Consistently high

Example:
Training a flare prediction model from scratch requires massive labeled dataset. With Surya, fine-tuning on a few thousand examples achieves better performance.

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7. LIMITATIONS AND FUTURE DIRECTIONS

7.1 Current Limitations

1. Computational Requirements

- Model size: 366M parameters (~1.4GB)
- Inference: Requires GPU for real-time prediction
- Training: Needs 128 GPUs for full training from scratch

2. Data Dependency

- Requires continuous SDO data stream
- Performance degrades during SDO data gaps
- Limited to SDO instrument capabilities

3. Interpretability

- Learned features less interpretable than physics models
- Difficult to understand failure modes
- Black-box decision making for critical applications

4. Temporal Coverage

- Trained on 2011-2019 data (solar cycle 24)
- Performance on different solar cycle phases uncertain
- Extreme events (>X10 flares) underrepresented in training

5. Spatial Coverage

- Only observes Earth-facing solar hemisphere
- Cannot predict far-side events until they rotate into view
- ~13-day lag for far-side active regions

7.2 Future Research Directions

Near-Term (1-2 years):

1. **Multi-Mission Integration**

    - Incorporate STEREO spacecraft data (side views of Sun)
    - Add SDO/EVE spectral data
    - Integrate ground-based magnetogram networks

2. **Extended Prediction Horizons**

    - 48-hour flare forecasting
    - 7-day solar wind predictions
    - 27-day active region evolution (full solar rotation)

3. **Uncertainty Quantification**

    - Probabilistic predictions with confidence intervals
    - Ensemble methods for robust forecasting
    - Calibrated probability estimates

4. **Real-Time Deployment**

    - Integration with NOAA Space Weather Prediction Center
    - Automated alert systems
    - API for third-party applications

Long-Term (3-5 years):

1. **Physics-Informed ML**

    - Hybrid models combining Surya with MHD simulations
    - Physical constraint enforcement
    - Interpretable feature extraction

2. **Multi-Messenger Observations**

    - Radio burst prediction
    - Energetic particle forecasting
    - Coronal mass ejection (CME) arrival time prediction

3. **Generalization to Other Stars**

    - Apply Surya architecture to stellar observations
    - Exoplanet host star activity prediction
    - Stellar flare forecasting for habitable zone studies

4. **On-Board Spacecraft AI**

    - Deploy lightweight Surya variant on satellites
    - Autonomous space weather prediction
    - Reduce ground communication dependencies

7.3 SuryaBench: Standardizing Evaluation

Purpose:
Accompanying Surya's release, the team created **SuryaBench** - a standardized benchmark for heliophysics AI models.

Components:

1. Curated datasets for each task
2. Train/validation/test splits
3. Evaluation metrics and protocols
4. Baseline model implementations

Impact:

- Enables fair comparison between methods
- Accelerates research by providing common evaluation framework

- Lowers barrier to entry for new researchers
- Facilitates reproducibility

Available at:
https://huggingface.co/collections/nasa-impact/suryabench-68265ce306fc2470c121af7b

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8. BROADER SIGNIFICANCE

8.1 Scientific Impact

Paradigm Shift:
Surya demonstrates that modern AI foundation models can successfully tackle complex scientific problems, opening doors for:

- Climate science
- Oceanography
- Seismology
- Planetary science
- Astrophysics

Quote from IBM Research:
"Surya is part of a broader effort at IBM to embrace generative and automated approaches that empower algorithms to be discovered, tested, and evolved at scale. Surya is the most recent example of how IBM sees AI not just as a tool, but as an originator and driver of scientific discovery."

8.2 Societal Impact

Economic Protection:

- Prevents billions in infrastructure damage
- Enables continued operation during solar storms
- Protects critical supply chains (satellites, GPS, power)

Safety Enhancement:

- Reduces astronaut radiation exposure
- Protects airline passengers and crew
- Enables safer space exploration

Space Exploration Enablement:

- Critical for Mars missions (deep space radiation)
- Lunar base planning (no magnetosphere protection)
- Long-duration spaceflight safety

8.3 Technological Heritage

Building on IBM-NASA Collaboration:

Surya follows IBM and NASA's successful Prithvi family of Earth observation AI models:

- **Prithvi-100M:** General-purpose Earth observation model
- **Prithvi-Weather:** Weather and climate forecasting
- **Prithvi-EO:** Multi-spectral Earth imaging

Shared Philosophy:

- Foundation models for scientific domains
- Open-source for research community
- Transfer learning for efficiency
- Physics-aware AI architectures

Quote from Juan Bernabé-Moreno (IBM Director):
"We've been on this journey of pushing the limits of technology with NASA since 2023, delivering pioneering foundational AI models to gain an unprecedented understanding of our planet Earth. With Surya we have created the first foundation model to look the Sun in the eye and forecast its moods."

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9. DEPLOYMENT AND ACCESS

9.1 Open Science Commitment

Availability:

- **Code:** Fully open-source on GitHub under Apache 2.0 license
- **Model weights:** Available on HuggingFace
- **Documentation:** Comprehensive tutorials and examples

- **Community:** Active development and support

Why Open Source?

1. Accelerates scientific research
2. Enables global collaboration
3. Allows independent validation
4. Democratizes access to advanced AI

9.2 Integration Pathways

Option 1: Direct Use via HuggingFace

```
from transformers import AutoModel

model = AutoModel.from_pretrained("nasa-ibm-ai4science/Surya-1.0")

# Run inference on SDO data
```

Option 2: Through IBM TerraTorch

- Streamlined fine-tuning pipeline
- Pre-configured for common tasks
- Optimized for deployment

Option 3: SuryaBench Framework

- Standardized evaluation
- Benchmarking against baselines
- Ready-to-use datasets

9.3 Computational Requirements

Inference (Prediction):

- **Minimum:** NVIDIA V100 GPU (16GB VRAM)
- **Recommended:** NVIDIA A100 GPU (40GB VRAM)
- **Latency:** ~2 seconds per prediction on A100
- **Throughput:** ~1800 predictions/hour

Fine-Tuning:

- **Minimum:** 4× NVIDIA A100 GPUs
- **Duration:** 2-24 hours depending on task
- **Data:** Task-specific dataset (1,000+ samples)

Full Training (from scratch):

- **Requirements:** 128× NVIDIA A100 GPUs
- **Duration:** ~2 weeks
- **Data:** Full SDO archive (257TB)
- **Not recommended:** Use pre-trained weights instead

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10. CONCLUSION

10.1 Summary of Key Points

What Surya Is:

- First foundation model for heliophysics
- 366M parameter vision transformer
- Trained on 9 years of full-resolution SDO data

Primary Purposes:

1. **Space weather forecasting** - 2-hour solar flare warning
2. **Scientific data analysis** - Automated feature extraction from petabytes
3. **Foundation for space physics AI** - Reusable model for multiple tasks

Main Functions:

1. **Solar flare forecasting** - 24-hour prediction with 2-hour lead time
2. **Active region segmentation** - Automatic AR identification and tracking
3. **EUV spectra prediction** - 1343-dimensional irradiance forecasting
4. **Solar wind forecasting** - 4-day ahead speed prediction

Key Applications:

- Astronaut and spacecraft protection
- Power grid stability

- Satellite operations
- Aviation safety
- GPS/navigation reliability
- Scientific research acceleration

10.2 Transformative Potential

Current State:
Surya is already being used by researchers worldwide for space weather studies and operational forecasting development.

Near Future (1-3 years):

- Integration with NOAA operational forecasting
- Real-time alert systems for critical infrastructure
- Widespread adoption by space agencies and satellite operators

Long-term Vision (5-10 years):

- Standard tool for all space weather operations
- Hybrid physics-AI forecasting systems
- Application to stellar activity prediction
- Foundation for deep space exploration planning

10.3 Final Perspective

Surya represents a fundamental advance in our ability to predict and respond to solar activity. By combining:

- **Massive observational datasets** (15 years of SDO)
- **Modern AI architectures** (vision transformers)
- **Domain expertise** (collaboration with leading heliophysicists)

...the project has created a tool that both enables better space weather forecasting today and opens new pathways for scientific discovery tomorrow.

Quote from Dr. Andrés Muñoz-Jaramillo:
"Our hope is that the model has learned all the critical processes behind our star's evolution through time so that we can extract actionable insights."

This hope is being realized as Surya moves from research prototype to operational tool, protecting humanity's increasingly space-dependent civilization.

REFERENCES

Primary Sources:

1. IBM Research Blog (2024). "Here comes a foundation model for the Sun." https://research.ibm.com/blog/surya-heliophysics-ai-model-sun

2. Roy et al. (2024). "Surya: Foundation Model for Heliophysics." arXiv:2508.14112. https://arxiv.org/abs/2508.14112

3. NASA-IMPACT GitHub Repository. "Surya: Implementation of the Surya Foundation Model." https://github.com/NASA-IMPACT/Surya

4. HuggingFace Model Hub. "nasa-ibm-ai4science/Surya-1.0" https://huggingface.co/nasa-ibm-ai4science

Supporting Documentation:

5. SuryaBench Benchmark Suite. HuggingFace Datasets. https://huggingface.co/collections/nasa-impact/suryabench-68265ce306fc2470c121af7b

6. IBM TerraTorch Documentation. https://ibm.github.io/terratorch/

APPENDIX: TECHNICAL GLOSSARY

Active Region (AR): Area of strong magnetic field on the Sun's surface, source of solar flares and CMEs.

AIA (Atmospheric Imaging Assembly): SDO instrument imaging Sun in extreme ultraviolet wavelengths.

Coronal Mass Ejection (CME): Large expulsion of plasma and magnetic field from Sun's corona.

Foundation Model: Large pre-trained AI model that can be adapted for multiple downstream tasks.

Geomagnetic Storm: Disturbance in Earth's magnetosphere caused by solar wind.

HMI (Helioseismic and Magnetic Imager): SDO instrument measuring Sun's magnetic field.

IoU (Intersection over Union): Metric for segmentation accuracy, ranges 0-1.

L1 Point: Lagrange point 1.5 million km from Earth where solar wind monitors are stationed.

MAPE (Mean Absolute Percentage Error): Metric for prediction accuracy in regression tasks.

SDO (Solar Dynamics Observatory): NASA spacecraft continuously observing Sun since 2010.

Solar Flare: Sudden burst of electromagnetic radiation from Sun's surface.

Solar Wind: Stream of charged particles flowing from Sun throughout heliosphere.

TSS (True Skill Statistic): Metric for binary classification accounting for class imbalance, ranges -1 to +1.

Vision Transformer: Neural network architecture using attention mechanisms for image processing.

END OF DOCUMENT

Prepared for review documentation